

Genetics and inheritance

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Competency: The learner appreciates that transmission of characteristics is from parents to offspring through a process known as heredity.

Inheritance (or heredity) is the process by which genetic information is passed down from parents to their offspring. The scientific study of inheritance is called genetics. We can observe inheritance in the resemblances between family members, such as sharing eye colour, height, or even certain medical conditions.

The language of genetics (key terminology)

To understand inheritance, we must first learn its specific language.

1. Chromosome: A thread-like structure located inside the nucleus of a cell, composed of a long molecule of DNA (Deoxyribonucleic Acid). Chromosomes are the carriers of genetic information. Humans have 46 chromosomes, arranged in 23 pairs.
2. Gene: A specific segment of DNA on a chromosome that controls a particular trait. For example, there is a gene for tongue rolling, a gene for earlobe attachment, and a gene for producing the pigment melanin.
3. Allele: The different forms or variations of a single gene. For instance, the gene for height in pea plants has two alleles: one for tallness (T) and one for shortness (t). For every gene, an individual inherits one allele from each parent.
4. Dominant allele: An allele that is always expressed in the physical appearance (phenotype) of an organism, even if only one copy is present. It is represented by a capital letter (e.g., T for tallness).
5. Recessive allele: An allele that is only expressed in the phenotype if two copies are present (i.e., when the dominant allele is absent). It is represented by a lowercase letter (e.g., t for shortness).
6. Genotype: The genetic makeup of an organism for a particular trait, represented by its combination of alleles. Examples: TT, Tt, tt.
7. Phenotype: The observable physical or biochemical characteristic of an organism, which is determined by its genotype. Examples: 'Tall plant', 'Short plant', 'Blood group A'.
8. Homozygous: An individual with two identical alleles for a particular trait.
 - a) Homozygous dominant: Genotype is TT. The phenotype is dominant.
 - b) Homozygous recessive: Genotype is tt. The phenotype is recessive.

9. Heterozygous: An individual with two different alleles for a particular trait. The genotype is Tt. The phenotype is dominant. A heterozygous individual is often called a carrier for the recessive trait.

Monohybrid inheritance and genetic crossings

Monohybrid inheritance is the study of how a single gene (controlling one characteristic) is passed from one generation to the next. We can predict the outcomes of a genetic cross.

Mendelian genetics



26.1: Gregor Mendel (1822–1884)

[Gregor Mendel](#) (1822-1884) was an Austrian Augustinian monk and scientist, known as the "[Father of modern genetics](#)" for his pea plant experiments that revealed the fundamental laws of inheritance, introducing concepts like dominant/recessive traits and how traits are passed as discrete units (genes) from parents to offspring.

His work laid the groundwork for genetics, though it was largely unrecognized until the early 1900s, long after his death.

Mendelian monohybrid inheritance with pea plants demonstrates how single traits are passed down, showing dominance and recessiveness through controlled crosses ([P \(parental\)](#), [F₁](#), [F₂](#) generations). Mendel crossed true breeding plants (e.g., tall crossed with short/dwarf), resulting in all dominant F₁ hybrids (e.g., all tall). Self-pollinating the F₁ generation yielded an F₂ with a characteristic 3 dominant: 1 recessive phenotypic ratio, revealing hidden recessive traits and supporting his laws of dominance and segregation.

Key concepts and steps

1. [True-breeding parents \(parental generation\)](#)

Mendel started with plants that always produced offspring with the same trait (e.g., pure tall and pure short).

He accurately removed anthers to prevent self-pollination before controlled cross-pollination.

2. [First filial generation \(F₁\)](#)

Crossing the pure parents (e.g., Tall x Dwarf) resulted in 100% hybrid offspring that showed only the dominant trait (e.g., all tall).

This demonstrated the Law of Dominance, where one allele (dominant) masks the other (recessive).

3. [Second filial generation \(F₂\)](#)

- Mendel self-pollinated the **F₁** hybrids (e.g., tall x tall).
- The **F₂** generation showed both traits reappearing, with a consistent 3 (dominant): 1 (recessive) phenotypic ratio (e.g., 3 tall: 1 dwarf/short).
- ❖ This confirmed the **Law of Segregation**, where alleles separate during gamete formation, allowing recessive traits to reappear.

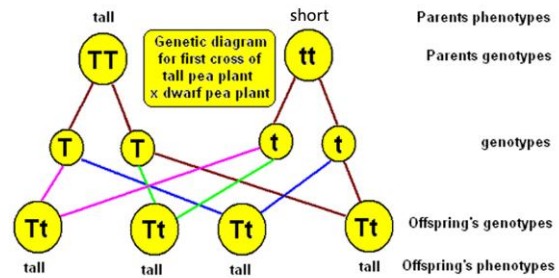
Example: Stem height in garden pea plants

Parental:

Homozygous tall (TT) x

Homozygous dwarf/short (tt)

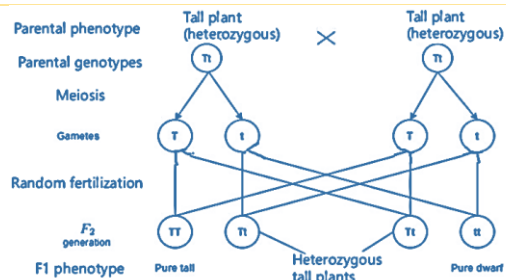
F₁: All heterozygous tall (Tt)



F₁ Self-cross (Tt x Tt)/ selfing the F₁ generation offspring

Genotypic ratio: 1TT : 2Tt : 1tt

Phenotypic ratio: 3Tall : 1dwarf



Why did Mendel choose the pea plants?

- Easy to cross, controlled pollination is simple and possible.*
- Pure breeding lines for specific traits were easy to establish (dominant and recessive traits).*
- Fast life cycles enabling multiple generations could be studied quickly.*
- Contrasting traits which were the clear, distinct forms for each trait (for example; tall/ short, purple/ white flowered, round/ wrinkled seeds).*

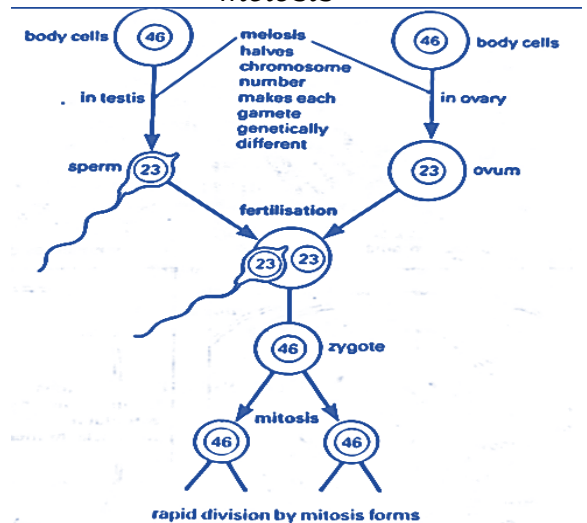
Inheritance of human traits and genetic disorders

How traits are transmitted

During sexual reproduction, the father's sperm cell and the mother's egg cell each contribute 23 chromosomes. When these combine during fertilization, they form a zygote with 46 chromosomes. This zygote then grows into a baby, carrying traits from both parents.

If both parents carry a recessive gene for a condition (like sickle cell anaemia), their child has a chance of inheriting the condition. If only one parent has the gene, the child may become a carrier without showing symptoms.

How gametes are formed from meiosis



a) Albinism

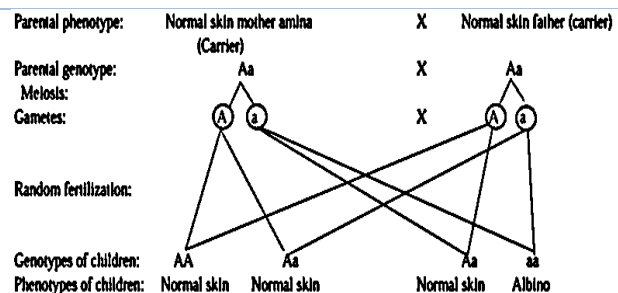
Albinism is a genetic condition where the body is unable to produce the pigment melanin. It is caused by a recessive allele (**a**). The dominant allele (**A**) codes for normal pigment production.

Genotypes and phenotypes

AA: Normal (Homozygous dominant)

Aa: Normal, but a carrier of the albinism allele (Heterozygous)

aa: Has albinism (Homozygous recessive)



A genetic cross where the two parents are carriers of albinism trait.



26.2: An albino

Task (Perform this genetic cross)

Two parents are carriers for albinism (Aa). What is the probability of them having a child with albinism?

Using the same genetic cross as in the case of the pea plant example, the probability of having a child with the genotype **aa** is **1 in 4 (or 25%)**.

Signs and symptoms of albinism/ manifestation

Albinism symptoms primarily involve lack of pigment in skin, hair, and eyes, leading to very pale skin that burns easily, white or light-coloured hair, and light eyes (blue, grey, or brown) with potential pinkish tint, alongside significant vision problems like light sensitivity (*photophobia*), [nystagmus](#) (*involuntary eye movement*), [strabismus](#) (crossed eyes), and [poor vision](#). Skin may also develop freckles, moles, or sunspots, and sometimes colour can increase with age.

b) Sickle cell anaemia

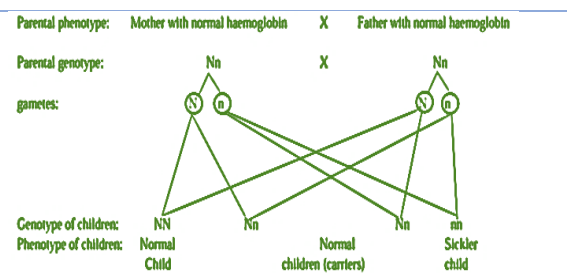
This is a genetic blood disorder caused by a recessive allele (**n**). The dominant allele (**N**) codes for normal, round red blood cells. Individuals with sickle cell anaemia have sickle-shaped red blood cells that can block blood vessels and carry less oxygen.

Genotypes and phenotypes:

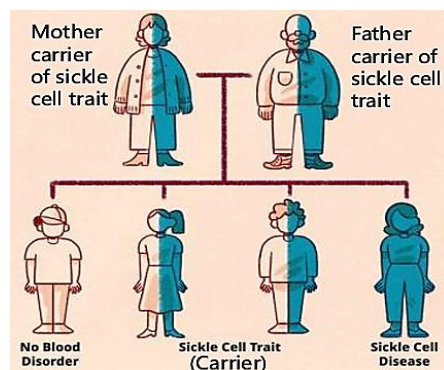
NN: Normal (Homozygous dominant)

Nn: Has "sickle cell trait" (Heterozygous). These individuals are carriers and are generally healthy but have resistance to malaria.

nn: Has sickle cell anaemia (Homozygous recessive).



A genetic cross where the two parents are carriers of sickle cell trait.

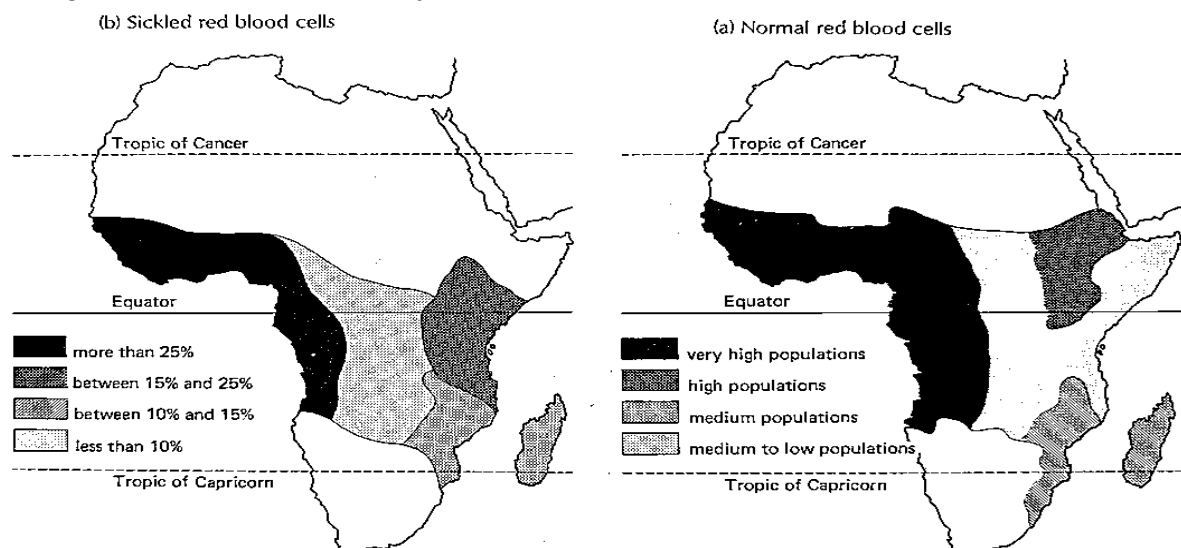


26.3: Family tree: Sickle cell anaemia

Symptoms of sickle cell anaemia disease

These include severe pain episodes (crises) in the chest, abdomen, arms, and legs; fatigue and paleness from anaemia; swollen hands and feet in infants (dactylitis); jaundice (yellowing skin/eyes); frequent infections; and potential complications like stroke, vision problems, and [acute chest syndrome](#), with symptoms often appearing after four months of age.

Geographical distribution of sickle cell anaemia and malaria in the tropics



A map showing the distribution frequency of people carrying the gene for sickle cell haemoglobin

A map showing the distribution frequency of *Anopheles* mosquitoes which act as vectors for the organisms causing malaria

ABO blood group system

This is an example of co-dominance and multiple alleles. There are three alleles: I^A , I^B , and i . I^A and I^B are co-dominant (both are expressed if present). i is recessive to both I^A and I^B .

Genotype(s)	Phenotype (blood group)
$I^A I^A$ or $I^A i$	A
$I^B I^B$ or $I^B i$	B
$I^A I^B$	AB
ii	O

Sex determination in humans

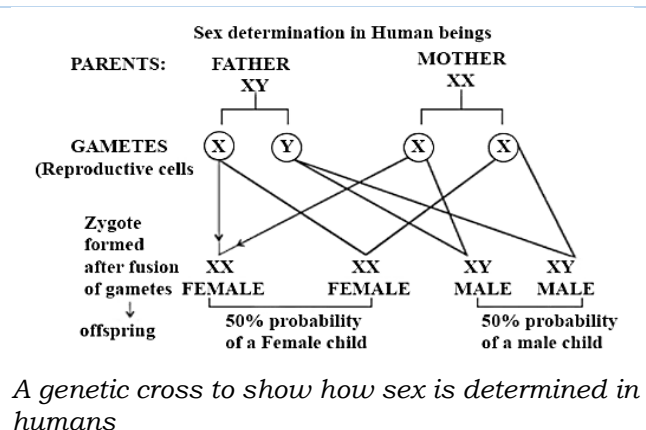
Sex determination is the biological process by which the sex of a baby (male or female) is decided at fertilisation.

Chromosomes Involved

Humans have 23 pairs of chromosomes
One pair are the sex chromosomes
Female: XX and Male: XY

How sex is determined

The mother's ovum always carries an X chromosome
The father's sperm carries either: X chromosome or Y chromosome



Note: Therefore, the father determines the sex of the child, not the mother. The chances of producing a boy or girl are equal (50%)

Sex linkage in humans

Sex linkage refers to traits or conditions whose genes are carried on the sex chromosomes, mainly the X chromosome.

Examples of sex-linked conditions

Condition	Chromosome involved	Effect
1. Colour blindness	X-linked	Difficulty distinguishing colours
2. Haemophilia	X-linked	Poor/ delayed blood clotting

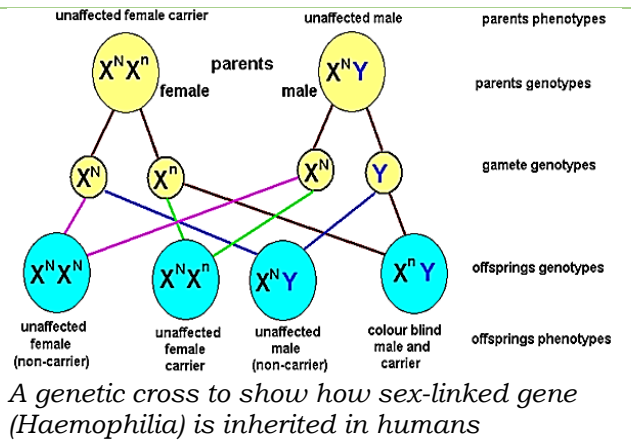
Genetic illustration showing how a boy (Brian) inherited colour blindness from his mother:

Let:

- X^N = normal vision allele (dominant)
- X^n = colour blindness allele (recessive)
- Y = male chromosome

Brian's mother is a carrier ($X^N X^n$)
 Brian's father has normal vision ($X^N Y$)

Cross: $X^N X^n \times X^N Y$



Brian inherited the X^n allele from his mother and Y from his father, making him $X^n Y$ (colour blind)

His mother shows no symptoms because she has a dominant normal allele (X^N) that masks the recessive allele

Why sex-linked traits are common in males

Males have only one X chromosome (XY)

A defective gene on the X chromosome is expressed immediately in males

Females (XX) may be carriers without showing the trait

Importance of understanding inheritance

1. Genetic counselling: It helps families understand the risks of inheriting genetic disorders and make informed decisions.
2. Agriculture (selective breeding): Farmers and breeders can select parent organisms with desirable traits (e.g., high milk yield in cows, disease resistance in crops) to produce improved offspring.
3. Paternity testing: DNA analysis based on inheritance patterns can be used to determine parentage.
4. Forensic science: DNA fingerprinting helps identify individuals at crime scenes.

TOPICAL ACTIVITIES

To test whether the learner is able to:

- a) understand the process of cell division and significance of meiosis
- b) understand the concept of inheritance using genetic diagrams
- c) understand and explain sex determination in humans
- d) understand and explain sex linkage in humans

Activity 26.1

Mr. and Mrs. Ocen's child, Abbo, was diagnosed with sickle cell anaemia.

Both parents appear healthy and have never shown symptoms. At the local hospital, a genetic counsellor explains how carrier parents can pass the

gene on to their child, using a genetic diagram, and highlights the role of meiosis in inheritance.

Tasks:

- a) Using a genetic illustration, show how the child (Abbo) could have acquired the genetic defect from the parents.**
- b) Explain the role of meiosis in passing on Abbo's genetic defect from carrier parents to their child.**
- c) Discuss two reasons why genetic testing and counselling are important for couples in Uganda before having children.**

Activity 26.2

Brian, a schoolboy, fails to distinguish red from green during science activities. His maternal uncle has the same condition. The teacher explains that the disorder is X-linked and is passed from carrier mothers to their sons.

Tasks:

- a) Using a genetic diagram, explain how Brian inherited colour blindness from his mother, who shows no symptoms.**
- b) Describe how sex determination and sex linkage are connected and why such conditions affect boys more than girls.**
- c) Explain the role of meiosis in forming gametes that carry sex-linked traits.**

Activity 26.3

A family in Soroti has three children; one was born with albinism. Both parents have normal skin. A health worker explains that it is possible for two parents with normal skin colour to bear a child with albinism

Tasks:

- a) Use a genetic diagram to show how two parents can have a child with albinism.**
- b) Explain how meiosis is involved in passing the albinism gene from both parents.**
- c) State two challenges people with albinism face in Uganda and suggest how schools and communities can support them.**

Activity 26.4 *(Integrated activity)*

At Arua Regional Referral Hospital, a baby requires an urgent blood transfusion. The mother's blood group is O, the father's is AB, and the baby is blood group O. The medical team is surprised. A student nurse suspects an error. The doctor explains blood group inheritance, using codominance (A and B) and recessiveness (O), and discusses how genetic testing resolves parentage disputes.

Tasks:

- a) Use your knowledge of genetics to show that a parent with blood group AB cannot have a child with blood group O and explain your**

reasoning.

b) Describe the inheritance of blood groups using codominance and recessiveness.

c) Explain how genetic testing can help resolve family disputes or confirm biological parentage in such cases.

Did you know?

- ✓ **Your Genetics is like a biological instruction book written in DNA just a tiny twist in the code can decide whether you have brown hair, dimples, or even resistance to certain diseases, making every person on Earth one of a kind!**
- ✓ **The survival of entire species can depend on tiny differences called Variation for example, a single mutation can make bacteria resistant to antibiotics or give some individuals a survival edge; through Natural Selection, these advantageous traits spread rapidly, meaning evolution is not slow and distant it is happening around us right now, shaping life in real time and determining which organisms thrive or disappear?**